

EXPERIMENT 3(C) - MULTIPLE REGRESSION ANALYSIS

DATASET

	Glucose	BloodPressure	SkinThickness	Insulin	DiabetesPedigreeFunction
0	182	73	37	215	1.879
1	131	98	34	153	1.422
2	172	74	48	133	1.183
3	94	67	42	276	0.285
4	186	73	18	161	1.269

	Age	BMI
0	69	43.56
1	48	36.55
2	52	38.34
3	47	28.54
4	40	38.54

Dataset Shape:
(100, 7)

DATASET INFORMATION

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Glucose                100 non-null   int32
1   BloodPressure          100 non-null   int32
2   SkinThickness          100 non-null   int32
3   Insulin                100 non-null   int32
4   DiabetesPedigreeFunction 100 non-null   float64
5   Age                   100 non-null   int32
6   BMI                   100 non-null   float64
dtypes: float64(2), int32(5)
memory usage: 3.6 KB
None
```

MISSING VALUES

```
Glucose      0
BloodPressure 0
SkinThickness 0
Insulin      0
DiabetesPedigreeFunction 0
Age          0
BMI          0
dtype: int64
```

VARIABLES

Independent Variables:

- Glucose
- BloodPressure
- SkinThickness
- Insulin
- DiabetesPedigreeFunction
- Age

Dependent Variable:

- BMI

TRAIN / TEST SPLIT

TRAIN / TEST SPLIT

Total samples : 100
Training samples : 80
Testing samples : 20

MODEL TRAINING

Multiple Linear Regression model trained successfully.

REGRESSION COEFFICIENTS

Feature	Coefficient
Glucose	0.059489
BloodPressure	0.022182
SkinThickness	0.015145
Insulin	0.010720
DiabetesPedigreeFunction	1.875004
Age	0.057943

Intercept:
16.4067

BMI PREDICTIONS

Actual BMI	Predicted BMI
35.86	35.10
23.84	27.88
36.37	35.98
39.49	36.13
31.40	33.16
35.74	33.69
31.86	34.42
33.92	35.50
28.52	35.89
43.56	39.24
32.82	32.04
35.76	32.26
29.23	31.20
30.67	31.46
33.80	34.27
38.54	35.67
29.44	28.05
32.30	31.26
32.45	34.25
37.34	35.56

MODEL PERFORMANCE

R^2 Score : 0.5890
Mean Squared Error : 7.6436
Mean Absolute Error : 2.2287

Actual vs Predicted BMI
Multiple Linear Regression



```
print("-" * 75)
print("EXPERIMENT 3(C) COMPLETED SUCCESSFULLY")
print("-" * 75)
```

